

WEB-IDS23

WEB-IDS23 is a network intrusion detection dataset that includes over 12 million flows, categorizing 20 attack types across FTP, HTTP/S, SMTP, SSH, and network scanning activities. This dataset is documented in the paper “Technical Report: Generating the WEB-IDS23 Dataset,” which provides insights into the generation, structure, and key characteristics of the dataset.

Data

The dataset is available as CSV files under x. Each file includes the data of one class, and each row corresponds to a flow extracted using Zeek FlowMeter. In total, the dataset includes over 12 million samples.

Labels

The dataset contains 21 class labels, representing 20 attack types and one benign class. The attack types can be categorized into five services, FTP, HTTP/S, SMTP, SSH and Miscellaneous which includes Portscan and Hostsweep.

Class Label	Description
benign	Normal network traffic with no attack activity.
bruteforce_http	Brute-force attack on an HTTP service using Hydra.
bruteforce_https	Brute-force attack on an HTTPS service using Hydra.
dos_http	Denial-of-service (DoS) attack on an HTTP service using sqlmap.
dos_https	Denial-of-service (DoS) attack on an HTTPS service using sqlmap.
ftp_login	Brute-force attack on an FTP service using Metasploit (auxiliary/scanner/ftp/ftp_login).
ftp_version	FTP service fingerprinting attack using Metasploit (auxiliary/scanner/ftp/ftp_version).
hostsweep_Pn	Host discovery scan using Nmap (-Pn flag).

Class Label	Description
hostswweep_sn	Host discovery scan using Nmap (-sn flag).
portscan	Port scanning attack using Nmap (-sS flag).
revshell_http	Reverse shell exploit over HTTP using Selenium WebDriver and netcat.
revshell_https	Reverse shell exploit over HTTPS using Selenium WebDriver and netcat.
smtp_enum	Enumeration of SMTP users using Metasploit (auxiliary/scanner/smtp/smtp_enum).
smtp_version	SMTP service fingerprinting attack using Metasploit (auxiliary/scanner/smtp/smtp_version).
sql_injection_http	SQL injection attack over HTTP using Selenium WebDriver, python-requests, and sqlmap.
sql_injection_https	SQL injection attack over HTTPS using Selenium WebDriver, python-requests, and sqlmap.
ssh_login	Brute-force attack on an SSH service using Metasploit (auxiliary/scanner/ssh/ssh_login).
ssh_login_successful	Successful brute-force login on an SSH service using Metasploit (auxiliary/scanner/ssh/ssh_login).
ssrf_http	Server-side request forgery attack over HTTP using Selenium WebDriver.
ssrf_https	Server-side request forgery attack over HTTPS using Selenium WebDriver.

Class Label	Description
xss_http	Cross-site scripting (XSS) attack over HTTP using Selenium WebDriver.
xss_https	Cross-site scripting (XSS) attack over HTTPS using Selenium WebDriver.

Features

The dataset includes 82 flow-level features, which capture various aspects of network traffic. The majority of the features are extracted using the Zeek FlowMeter tool. Additionally, the dataset includes information about the service type, the traffic direction and finally a fine-grained class label.

Feature Name	Source	Description
uid	Zeek	The ID of the flow as given by Zeek.
ts	Zeek	Timestamp of the flow.
id.orig_h	Zeek	Originating host IP address.
id.resp_h	Zeek	Responding host IP address.
service	Zeek	The service associated with the flow, e.g., <code>http</code> or <code>ssl</code> .
flow_duration	Zeek FlowMeter	The length of the flow in seconds (max precision ms). If only one packet was seen, the duration is 0.
fwd_pkts_tot	Zeek FlowMeter	The number of packets traveling in the forward direction.
bwd_pkts_tot	Zeek FlowMeter	The number of packets traveling in the backward direction.

Feature Name	Source	Description
fwd_data_pkts_tot	Zeek FlowMeter	The number of packets traveling in the forward direction that have a payload.
bwd_data_pkts_tot	Zeek FlowMeter	The number of packets traveling in the backward direction that have a payload.
fwd_pkts_per_sec	Zeek FlowMeter	The average number of forward packets transmitted per second during the flow. If the duration is 0, this feature is also set to 0.
bwd_pkts_per_sec	Zeek FlowMeter	The average number of backward packets transmitted per second during the flow. If the duration is 0, this feature is also set to 0.
flow_pkts_per_sec	Zeek FlowMeter	The average number of packets transmitted per second during the flow. If the duration is 0, this feature is also set to 0.

Feature Name	Source	Description
down_up_ratio	Zeek FlowMeter	The number of backward packets divided by the number of forward packets. If the number of forward packets is 0, this feature is also set to 0.
fwd_header_size_tot	Zeek FlowMeter	The total number of bytes in the headers of forward packets.
fwd_header_size_min	Zeek FlowMeter	The smallest header size among forward packets.
fwd_header_size_max	Zeek FlowMeter	The largest header size among forward packets.
bwd_header_size_tot	Zeek FlowMeter	The total number of bytes in the headers of backward packets.
bwd_header_size_min	Zeek FlowMeter	The smallest header size among backward packets.
bwd_header_size_max	Zeek FlowMeter	The largest header size among backward packets.
flow_FIN_flag_count	Zeek FlowMeter	The total number of FIN flags seen in a TCP flow. If the flow is not TCP, this feature is set to 0.
flow_SYN_flag_count	Zeek FlowMeter	The total number of SYN flags seen in a TCP flow. If the flow is not TCP, this feature is set to 0.

Feature Name	Source	Description
flow_RST_flag_count	Zeek FlowMeter	The total number of RST flags seen in a TCP flow. If the flow is not TCP, this feature is set to 0.
fwd_PSH_flag_count	Zeek FlowMeter	The total number of PSH flags seen in the forward direction of a TCP flow. If the flow is not TCP, this feature is set to 0.
bwd_PSH_flag_count	Zeek FlowMeter	The total number of PSH flags seen in the backward direction of a TCP flow. If the flow is not TCP, this feature is set to 0.
flow_ACK_flag_count	Zeek FlowMeter	The total number of ACK flags seen in a TCP flow. If the flow is not TCP, this feature is set to 0.
fwd_URG_flag_count	Zeek FlowMeter	The total number of URG flags seen in the forward direction of a TCP flow. If the flow is not TCP, this feature is set to 0.
bwd_URG_flag_count	Zeek FlowMeter	The total number of URG flags seen in the backward direction of a TCP flow. If the flow is not TCP, this feature is set to 0.

Feature Name	Source	Description
flow_CWR_flag_count	Zeek FlowMeter	The total number of CWR flags seen in a TCP flow. If the flow is not TCP, this feature is set to 0.
flow_ECE_flag_count	Zeek FlowMeter	The total number of ECE flags seen in a TCP flow. If the flow is not TCP, this feature is set to 0.
payload_bytes_per_second	Zeek FlowMeter	The average number of payload bytes transmitted per second. If the duration is 0, this feature is also set to 0.
fwd_init_window_size	Zeek FlowMeter	The initial window size in bytes of the first packet in the forward direction.
bwd_init_window_size	Zeek FlowMeter	The initial window size in bytes of the first packet in the backward direction.
fwd_last_window_size	Zeek FlowMeter	The window size in bytes of the last packet in the forward direction.
bwd_last_window_size	Zeek FlowMeter	The window size in bytes of the last packet in the backward direction.
traffic_direction	Set using IP addresses	Indicates the direction of traffic in the network, e.g., client -> server .

Feature Name	Source	Description
attack	Set using logs	Binary indicator of whether the flow is an attack.
attack_type	Set using logs	The fine-grained label.

Citation

If you use this dataset in your research, please cite the following paper:

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